

Sumit Kumar Chaudhary

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EDUCATION

Year	Degree/Certificate	Institute	CPI/Percent
2022-2024	M.Tech CSE	Indian Institute of Technology, Kanpur	7.71/10
2017-2021	B.Tech CSE	MMMUT, Gorakhpur	7.47/10
2016	CBSE (XII)	Chinmaya Vidyalaya , Bokaro	86.6 %
2014	ICSE(X)	De Nobili School C.T.P.S. , Bokaro	93.17%

PROFESSIONAL EXPERIENCE

- SDE-1 | C3iHub, IIT-KANPUR** Jul'24 - Present
 - Engineered a **Generative AI tool for cybersecurity assessment**, implementing a RAG pipeline with semantic search and cosine similarity to analyze unstructured logs and policy documents.
 - Architected the AI tool as a **containerized microservice** and developed a suite of REST APIs to integrate its capabilities into a standalone desktop application.
 - Designed the **end-to-end data workflow**, defining API contracts for document ingestion, real-time processing status, and the delivery of structured AI-generated insights to the end-user.
- Deep Learning Intern | Center for Developing Intelligent Systems(CDIS) IIT-KANPUR** Jan'24 - Mar'24
 - Developed a Deep Learning model to determine pediatric bone age from X-ray images, achieving a **Mean Average Error of 3 months**
 - Integrated the model with a NodeJS backend and web interface to provide real-time predictions from uploaded X-rays

PROJECTS

- Predictive Modeling for High-Dimensional Binary Classification**
 - Solved a complex binary classification challenge by building a custom SVM solver using primal gradient descent.
 - Developed advanced feature mapping techniques to model the behavior of a non-linear physical system from challenge-response data.
 - Achieved **99% accuracy** on 20k test set in predicting XOR-PUF responses, demonstrating a robust method to mitigate tampering and cloning risks
- End-to-End OCR Pipeline for CAPTCHA Solving**
 - Built a complete Optical Character Recognition (OCR) pipeline involving image preprocessing (noise reduction, binarization), character segmentation using contour detection, and classification.
 - Explored character recognition using **SVM** and **Logistic Regression** and achieved **98.5%** accuracy with a compact 174KB model and quick (4ms) prediction time
- DeceptiSense: Cyber Attack Pattern Recognition (M.Tech Thesis)**
 - Developed an automated tool to recognize attack patterns in Dockerized honeypots, integrating the MITRE ATT&CK framework.
 - Built a malware-capturing pipeline with a vulnerable web app for real-time TTP mapping and incident response.
- CSCMM Cybersecurity Assessment Tool**
 - Led a 2-person team to build a full-stack Cybersecurity Maturity assessment platform for Govt. of India Project using ReactJS, NodeJS, and PostgreSQL.
 - Implemented weighted scoring logic, encrypted data handling, and automated data extraction from SQLite to CSV.

TECHNICAL SKILLS

- AI/ML/NLP:** Scikit-learn, Numpy , Pandas, Matplotlib , OpenAI API, Semantic Search, Text Embeddings,Generative AI, Deep Learnig , Exploratory Data Analysis, PyTorch
- Programming & Web:** FastAPI, REST APIs, Python, C++, React.js, Node.js, HTML, Tailwind CSS
- Databases & DevOps:** PostgreSQL, MongoDB, MySQL , Microservice Architecture, Docker, Git, CI/CD, AWS(S3 , EC2)
- Cyber Security :** Metasploit, MITRE ATT&CK, NIST CSF , Wireshark, Burp Suite